Evaluation

For the evaluation we tested the controller on the pendulum model in Simulink using different initial angles. The main goal was to investigate how the initial state affects the ability of the controller to stabilize the pendulum and whether stabilization is possible for all conditions. The actuator limits play a role because they determine the maximum torque that the controller can apply. If the controller requests a torque that exceeds these limits the actuator. The actual applied torque is limited. Therefore the controller may be able to stabilize some conditions but fail for others when the available torque is insufficient.

The gravitational torque is one of the factors that affects the required control effort and is given by

$$\tau_g = mgl\sin(\theta)$$

where theta's the pendulum angle. For the parameters used in the model the maximum gravitational torque is 4.905 Nm

The gravitational torque reaches its magnitude around 90 degrees and decreases again as the angle approaches 180 degrees. Therefore a larger initial angle does not necessarily mean that a larger gravitational torque has to be overcome. The required control effort also depends on the motion of the pendulum and the response of the controller.

To test this we decided to limit the actuators to umin=−2.3 Nm and umax=2.3 Nm

These limits are below the gravitational torque of 4.905 Nm, providing a meaningful constraint on the controller. The actuator limits were then kept constant for all tests while only the initial angle was changed. This allows the effect of the angle to be examined without changing the available control torque.

Initial Angle	Actuator Limits	Stabilizes?
30	−2.3, 2.3	yes
60	−2.3, 2.3	yes
90	−2.3, 2.3	no
120	−2.3, 2.3	no
150	−2.3, 2.3	yes
180	−2.3, 2.3	yes

The results show that the controller was able to stabilize the pendulum for angles of 30 degrees, 60 degrees, 150 degrees and 180 degrees but failed to stabilize it for 90 degrees and 120 degrees. Therefore it is not always possible to stabilize the system from every tested initial state when the actuator is subject to these limits.

The results also show that the relationship between the angle and stabilization is not simply increasing or decreasing. In particular the controller failed at 90 degrees and 120 degrees. It was able to stabilize the pendulum again at 150 degrees and 180 degrees. This can be related to the gravitational torque, which increases towards 90 degrees and decreases again beyond 90. At 150 degrees for example the gravitational torque is lower than at 90 degrees despite the initial angle. Thus the initial angle alone does not determine whether the controller can stabilize the pendulum.

Overall the experiments demonstrate that the controller can stabilize the pendulum for some initial states but not for all of the tested initial states under the fixed actuator limits. The actuator saturation and the nonlinear relationship, between the pendulum angle and gravitational torque both influence whether stabilization is successful.